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Final Report

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Smart Waste Management System (Smart City)

1.0 Introduction

This study presents a Smart Waste Management System designed to enhance urban sanitation through the integration of Design Thinking and Artificial Intelligence. By utilizing Knowledge Representation (KR) and First-Order Logic (FOL), the system encodes essential rules to interpret real-time data, allowing it to recognize bin fill levels, predict waste accumulation, and prioritize collection tasks. This foundational logic is operationalized through a State Space Search model, which maps the sequential decision-making process from environmental sensing to operational execution, establishing a traceable path for identifying urgent bins and optimizing collection routes.

To facilitate autonomous interaction with the urban infrastructure, the system is defined as a Goal-Based Agent using the PEAS (Performance, Environment, Actuators, Sensors) framework. The agent perceives its environment through sensors, such as ultrasonic and weight detectors, and acts upon it using actuators like route guidance systems to maximize collection efficiency. By integrating these components, the proposed solution replaces inefficient fixed-schedule collections with a data-driven approach, ultimately reducing operational costs, minimizing fuel consumption, and promoting environmental sustainability.

2.0 Problem Statement

In many urban areas, waste bins often overflow because they are not emptied at the right time. This situation leads to unpleasant odors, attracts pests and creates an unsightly environment that affects residents' health and quality of life. The root cause is the use of fixed waste collection schedules that do not match the actual usage of each bin. Some bins may fill up quickly while others remain half-empty, causing inefficiency in the waste management process.

City authorities face challenges in monitoring thousands of bins across different locations, making it difficult to determine when and where waste collection is urgently needed. Therefore, there is a need for an intelligent system that can predict and manage waste collection more effectively to maintain city cleanliness and operational efficiency.

3.0 AI Solution

The proposed Smart Waste Management System uses Artificial Intelligence (AI), Internet of Things (IoT) and data-driven prediction models to transform how cities manage waste. Unlike traditional smart-bin approaches that only detect fill levels, our solution introduces multi-layer intelligence, enabling the system to think, predict, optimize and handle waste collection with minimal human intervention. The details of our multi-layer intelligence are as follows;

1. Multi-Sensor IoT Smart Bins - each waste bin is equipped with a combination of IoT sensors like ultrasonic sensors to detect fill height, weight sensors to measure load and temperature sensors to detect potential fire hazards.
2. Predictive Waste Generation AI Model - The system predicts when a bin will become full, how fast waste accumulates in different zones and studies waste management patterns based on time, events, holidays and others.
3. Autonomous Collection Recommendation System - When a bin is predicted to exceed 70%, the AI will automatically notify the nearest waste collection team

In addition to these features, the system uses AI to generate the most efficient vehicle routes in real time by considering the bin fill predictions, traffic conditions, fuel consumption and distance between each bins to ensure faster operations, reduced fuel usage and smarter allocation of city resources, making the entire waste management process more sustainable.

4.0 Goal

The main goal of this project is to maintain city cleanliness and sustainability by developing a Smart Waste Management System that uses Artificial Intelligence (AI) and Internet of Things (IoT) technologies to optimize waste collection schedules and routes. Specifically, the system aims to:

- **Optimize waste collection schedules using AI prediction.**

The system will collect real-time data from IoT sensors installed in waste bins to monitor their fill levels, weight, and disposal frequency. Using predictive analytics, AI will forecast when each bin is likely to reach full capacity and automatically schedule collection at the right time. This ensures that bins are emptied before overflowing, keeping the city clean and hygienic.

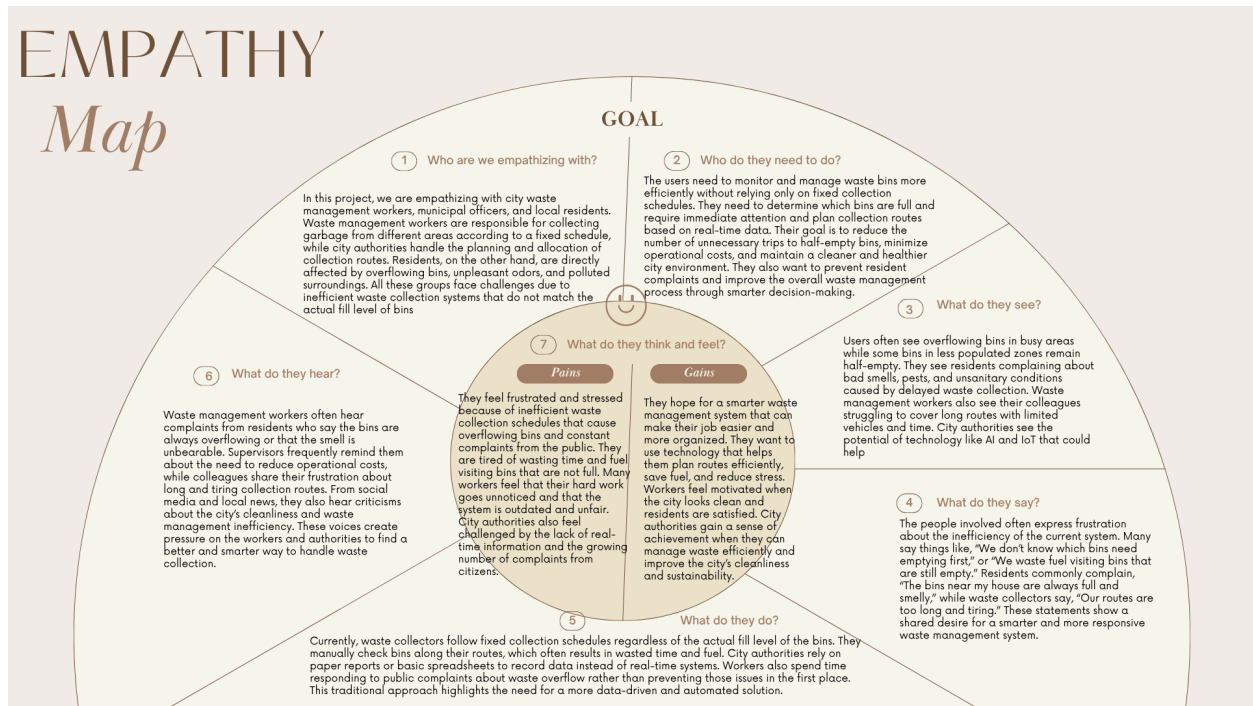
- **Reduce operational costs by minimizing unnecessary collection trips.**

Instead of following fixed collection schedules, the AI system will determine which bins actually require servicing and recommend the most efficient routes for waste collection vehicles. This reduces fuel consumption, vehicle wear, and manpower usage, leading to lower operational costs for city authorities.

- **Improve environmental sustainability by preventing waste overflow and pollution.**

By ensuring timely waste collection, the system helps prevent overflowing bins that cause unpleasant odors, attract pests, and lead to littering. In addition, optimized collection routes contribute to lower carbon emissions, supporting the city's goal of becoming cleaner, greener, and more sustainable.

5.0 Empathy



Link :

https://drive.google.com/drive/folders/1FeKkDQC0GD68LdYB88nfcESQ6USTing?usp=drive_link

6.0 Define

From the Empathize phase, we learned the city residents, waste collectors and municipal authorities experience challenges in maintaining efficient and timely waste collection. Residents complain about overflowing bins, unpleasant odors and the presence of pests that affect their health and environment. Waste collectors face difficulty planning routes efficiently because they have no real-time data on which bins are full. City authorities struggle to monitor thousands of bins across different zones without updated information.

6.1 Problem statement

The current waste management system is inefficient because waste bins are collected based on fixed schedules instead of actual usage which results in overflowing bins in some areas and wasted collection trips in others.

6.2 User needs

Understanding the needs of the main user groups involved in this system is important to support the cleanliness, efficiency of the organization operations and decision making. The user groups include residents, waste collection teams and city authorities.

6.2.1 Residents

Residents need cleaner surroundings and timely waste disposal to maintain hygiene and comfort. Overflowing bins can cause unpleasant odors and also health risks that can decrease the quality of life. This system helps ensure the resident lives with a safe and pleasant residential environment.

6.2.2 Waste collection teams

Waste collection teams need real-time information on bin status to plan efficient collection routes. This helps reduce unnecessary trips, save time and lower fuel consumption. This system helps to improve the operational efficiency and workload by providing easy access to the accurate data.

6.2.3 City authorities

City authorities need a centralized system that provides data-driven insights for better management and resource allocation. Systems that provide planning, performance tracking and

informed decision making are heavily needed for them. It enables efficient long term policy improvement and helps authorities make better decisions to improve waste management services across the city.

6.3 Insights

The key insight is that the waste collection process lacks real-time intelligence and predictive capability. This presents an opportunity to use AI and IoT to create a smarter, data-driven solution.

The proposed AI Smart Waste Management System will address this need by:

1. Collecting real-time data from IoT sensors on waste levels, weight and disposal frequency.
2. Predicting fill levels using AI algorithms to determine when each bin will need servicing.
3. Alerting waste collection teams automatically and suggesting the most efficient routes.

By redefining the problem and understanding user needs, our goal is to design an AI-based system that improves efficiency, cleanliness and sustainability in urban waste management.

7.0 Knowledge Representation

7.1 Knowledge Representation #1 Bin Fill-Level Knowledge

This KR stores knowledge about the waste bin's real-time status (fill level, weight, disposal frequency). It allows the AI system to recognize when a bin is almost full and requires attention.

FOL Representation

Let :

- $\text{Bin}(x)$ = x is a waste bin
- $\text{FillLevel}(x,y)$ = bin x has fill level y (percentage)
- $\text{NearFull}(x)$ - bin x is near full
- $\text{Threshold}(t)$ = threshold percentage (70%)

Natural Language : IF x is a bin AND the fill level of x is y AND the threshold is t AND y is greater than or equal to t THEN the bin x is considered near full.

$$\forall x \forall y \forall t [\text{Bin}(x) \wedge \text{FillLevel}(x,y) \wedge \text{Threshold}(t) \wedge y \geq t \rightarrow \text{NearFull}(x)]$$

If a bin reaches or exceeds the threshold (70%), the system marks it as 'near full.'

This KR is used by the predictive model to detect which bins need immediate collection. It helps the system maintain cleanliness and prevent overflow.

7.2 Knowledge Representation #2 Waste Accumulation Prediction Knowledge

This KR represents patterns of waste accumulation based on time, zone population, events, and historical data. The AI model uses this knowledge to predict when the bin will be full.

FOL Representation

Let :

- $\text{HighUsageZone}(z)$ = zone z produces waste quickly
- $\text{BinInZone}(x,z)$ = bin x is located in zone z
- $\text{PredictFull}(x,t)$ = bin x will be full at time t

Natural Language : IF x is a bin AND the bin is located in zone z AND zone z is a high waste usage zone THEN the system predicts that bin x will become full at time t.

$$\forall x \forall z \forall t [\text{Bin}(x) \wedge \text{BinInZone}(x,z) \wedge \text{HighUsageZone}(z) \rightarrow \text{PredictFull}(x,t)]$$

Bins in busy zones fill faster, so the system predicts earlier full times.

This KR allows the AI to forecast fill-level trends and schedule collection before overflow, helping achieve the goal of cleanliness and efficiency.

7.3 Knowledge Representation #3 Collection Priority Knowledge

This KR tells the system which bins should be collected first based on urgency.

FOL Representation

Let:

- NearFull(x) = bin x is near full
- Urgent(x) = bin x needs immediate collection
- EventDay(x) = today is a special event (more waste generated)

Natural Language :

1. IF bin x is near full THEN bin x becomes urgent :

$$\forall x [\text{NearFull}(x) \rightarrow \text{Urgent}(x)]$$

2. IF bin x has medium fill level AND today is an event day THEN bin x is urgent :

Let MediumFill(x) = 50-69%

$$\forall x \forall d [\text{MediumFill}(x) \wedge \text{EventDay}(d) \rightarrow \text{Urgent}(x)]$$

During events, waste increases so even medium-level bins become a high priority.

This KR allows the autonomous system to decide priority and reduce complaints by ensuring timely collection.

7.4 Knowledge Representation #4 Route Optimization Knowledge

This KR tells the system as AI needs knowledge of the distances, locations and travel time to generate the best route.

FOL Representation

Let :

- $\text{Bin}(x)$
- $\text{Distance}(x,y,z) = \text{distance between locations } x \text{ and } y \text{ is } d$
- $\text{Route}(a,b,r) = \text{route } r \text{ connects } a \text{ to } b$
- $\text{Shortest}(r) = r \text{ is the shortest route}$

Natural Language : IF r is a route from location a to location b AND route r is the shortest route THEN r is the optimal route.

$$\forall a \forall b \forall r [\text{Route}(a,b,r) \wedge \text{Shortest}(r) \rightarrow \text{OptimalRoute}(r)]$$

The AI must always select the shortest route for efficiency.

This KR helps reduce fuel usage, travel time and operational cost which directly supporting the project goal

7.5 Knowledge Presentation #5 Vehicle Assignment Knowledge

This KR tells the system about availability of vehicles and assigning the nearest one to an urgent bin.

FOL Representation

Let:

- $\text{Vehicle}(v)$
- $\text{Available}(v)$
- $\text{Near}(v, x)$
- $\text{Assign}(v, x)$

Natural Language: IF v is a vehicle AND v is available AND bin x is urgent AND vehicle v is near bin x THEN assign vehicle v to service bin x.

$$\forall v \forall x [\text{Vehicle}(v) \wedge \text{Available}(v) \wedge \text{Urgent}(x) \wedge \text{Near}(v, x) \rightarrow \text{Assign}(v, x)]$$

The nearest available truck must respond to urgent bins.

This KR reduces response time and ensures efficiency in waste handling.

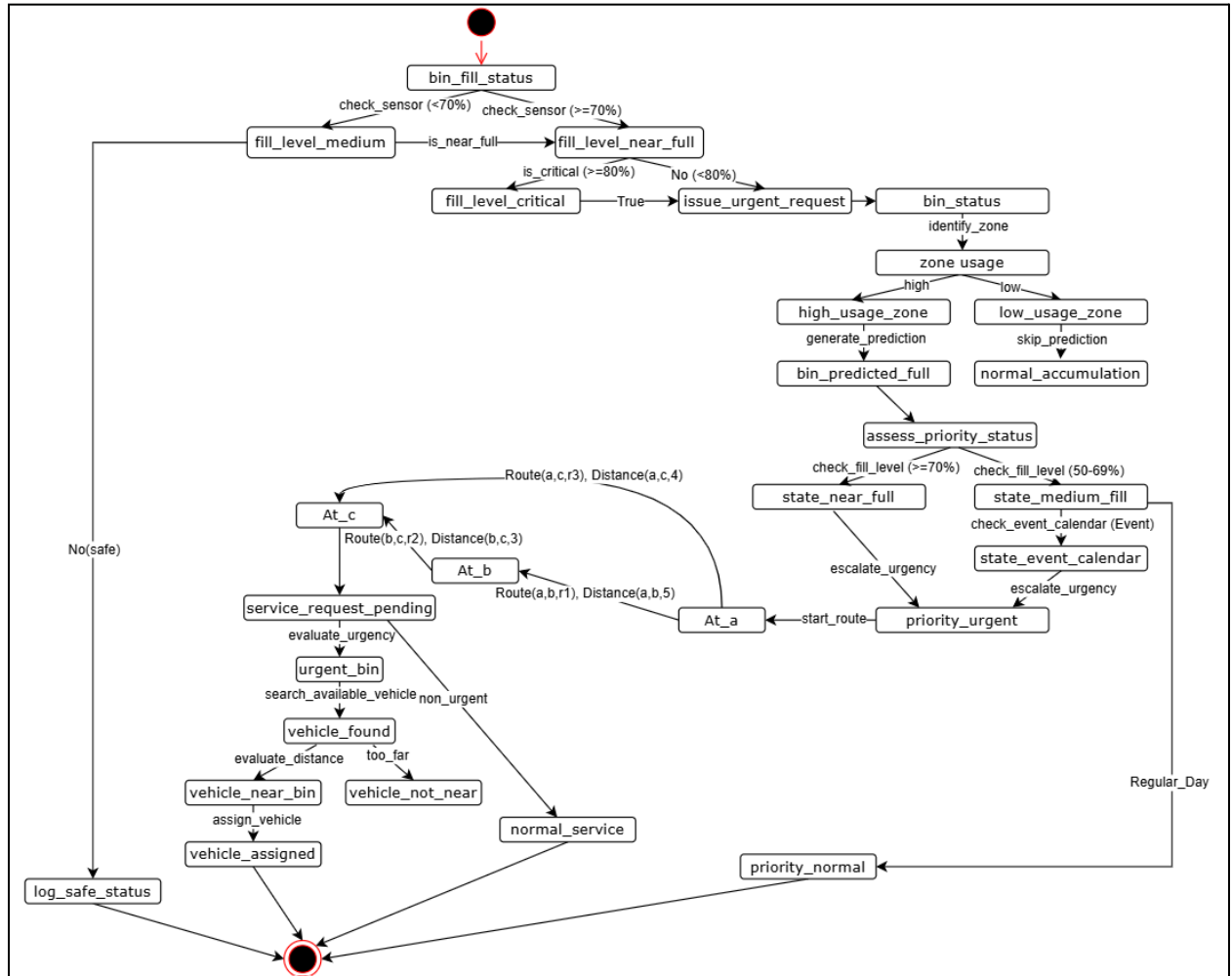
8.0 Supporting KR to The Project Goal

Table 1 shows the Knowledge Representations (KRs) that support the project goal by helping the AI optimize waste collection schedule, reduce operational cost and improve sustainability. KRs #1, #2, #3 ensure the system to detect full bins, predict waste levels and prioritize urgent bins to keep the city clean and optimize the collection schedule. Meanwhile, KRs #4 and #5 allow the AI to plan efficient routes and assign the right vehicles, which lowers operational costs. Altogether, KRs #1 until #5 work together to reduce overflow, minimize fuel usage and support long-term sustainability for city waste management.

Table 1 Supporting KR to the goals

Goal	Supporting KR
Optimize waste collection schedule	KR #1, #2, #3
Reduce operational cost	KR #4, #5
Improve sustainability	ALL

9.0 State Space Search



10.0 Explanation of States and Action

State (Nodes): Represents the current status

State	Description
bin_fill_status	The starting state where the system receives real time sensor data from the bin.
fill_level_medium	State where the fill level is below the NearFull threshold (< 70%)
fill_level_near_full	State where the fill level is $\geq 70\%$, corresponding to the NearFull(x) predicate from KR #1.
fill_level_critical	State where the fill level is $\geq 80\%$, requiring immediate collection.
issue_urgent_request	Goal State: The state where the AI signals the need for collection planning.
log_safe_status	Nothing happens where the system logs the data and returns to monitoring.
bin_status	The starting state where the system initiates the monitoring process for a specific bin.
zone_usage	State where waste generation rate of the bins is identified.
high_usage_zone	State where the bin is located in a zone that produces waste quickly.
low_usage_zone	State where the bin is located in a zone with normal waste generation.
bin_predicted_full	Goal state: The state where system predicts the bin will be full at future time t, based on high usage zone patterns, matching the PredictFull(x,t) predicate from KR #2.
normal_accumulation	End state: The state where the bin is expected to fill at a normal rate, no early prediction is generated, the bin returns to be monitored.
assess_priority_status	The starting state where the system initiates the priority evaluation process for a specific bin.
state_near_full	State where the bin's fill level is confirmed to be $\geq 70\%$, matching the NearFull(x) predicate.

state_medium_full	State where the bin's fill level is between 50% and 69%, requiring further context checks.
state_event_day	State where the system confirms that today is a special event day, implying higher waste generation.
state_regular_day	State where the system confirms today is not an event day.
priority_urgent	Goal State: The state where the bin is marked as "Urgent", triggering an immediate collection request.
priority_normal	End Path: The state where the bin is marked as standard priority, and no immediate alert is needed.
At_a	The lorry is currently at location a
At_b	The lorry is currently at location b
At_c	The lorry is currently at location c where Bin(c) is true
service_request_pending	State where a bin requires evaluation for collection.
urgent_bin	State where the bin is confirmed to require urgent service.
non_urgent_bin	State where the bin does not require urgent service.
vehicle_found	State where at least one vehicle is available.
vehicle_near_bin	State where an available vehicle is near the bin.
vehicle_not_near	State where no available vehicle is assigned to the bin.
vehicle_assigned	Goal State: The state where a vehicle is assigned to the bin.
normal_service	End State: The state where the bin is marked as standard service and no immediate service is needed.

Actions (Edges): Represent the AI's checks and status transitions.

Action	Description
check_sensor	Action to initiate reading and updating the current fill level.
is_near_full	Action to check if the level is $\geq 70\%$ (the threshold(t)).
is_critical	Action to check if the level is $\geq 80\%$ (a secondary, higher threshold).
Yes/True	Action for a condition check resulting in true.

No/False	Action for a condition check resulting in false.
identify_zone	Action to identify which zone the bin belongs to.
generate_prediction	Action to generate and predict early fullness.
skip_prediction	Action to log the bin as safe/non-urgent and continue normal monitoring.
high	Action for condition check resulting in the bin at a high usage zone.
low	Action for condition check resulting in the bin at a low usage zone.
check_fill_level	Action to read sensor data and categorize the bin as Low, Medium, or Near Full.
check_event_calendar	Action to query the system's database to verify if the current date is marked as EventDay(d).
escalate_urgency	Action to flag the bin as urgent based on the logic rules provided in KR #3.
maintain_normal	Actions to log the bin as safe/non-urgent and return to monitoring.
Route(a,b,r1), Distance(a,b,5)	A route r1 exists connecting location a to b. The distance between location a to b is 5.
Route(b,c,r2), Distance(b,c,3)	A route r2 exists connecting location b to c. The distance between location b to c is 3.
Route(a,c,r3), Distance(a,c,4)	A route r3 exists connecting location a to c. The distance between location a to c is 4.
evaluate_urgency	Action to determine if the bin is urgent to collect.
search_available_vehicle	Action to search for available vehicles.
evaluate_distance	Action to check distance between vehicle and bin.
assign_vehicle	Action to assign vehicle to bin
find_nearest	Action to find the nearest vehicle to the bin.
maintain_normal	Actions to log the bin as safe/non-urgent and return to monitoring.

11.0 Problem Formulation and Explanation

The combined state space problem is formulated as a single sequential decision process that enables the AI system to move from environmental sensing to operational execution.

Initial State

- **bin_fill_status**
The system begins by monitoring bin fill levels using real-time sensor data.

Actions and State Transitions

Let $S(x)$ be the set of action–state pairs that define valid transitions:

- $S(\text{bin_fill_status}) \rightarrow \{\text{fill_level_medium}, \text{fill_level_near_full}\}$
The system categorizes the bin based on a predefined threshold (70%).
- $S(\text{fill_level_near_full}) \rightarrow \{\text{issue_urgent_request}\}$
Near-full bins trigger further analysis.
- $S(\text{issue_urgent_request}) \rightarrow \{\text{zone_usage}\}$
The system evaluates waste accumulation patterns based on zone type.
- $S(\text{zone_usage}) \rightarrow \{\text{high_usage_zone}, \text{low_usage_zone}\}$
High-usage zones lead to early overflow prediction.
- $S(\text{high_usage_zone}) \rightarrow \{\text{bin_predicted_full}\}$
The system predicts future bin fullness.
- $S(\text{bin_predicted_full}) \rightarrow \{\text{assess_priority_status}\}$
Priority evaluation is initiated.
- $S(\text{assess_priority_status}) \rightarrow \{\text{priority_urgent}, \text{priority_normal}\}$
The system determines urgency based on fill level and context.
- $S(\text{priority_urgent}) \rightarrow \{\text{At}(a) \rightarrow \text{At}(c)\}$
Route optimization is performed to reach the bin via the shortest path.
- $S(\text{At}(c)) \rightarrow \{\text{service_request_pending} \rightarrow \text{vehicle_found} \rightarrow \text{vehicle_assigned}\}$
Vehicle availability is checked and the nearest vehicle is assigned.

Goal State

- vehicle_assigned
The system successfully assigns a collection vehicle to an urgent bin.

Path Cost

- The path cost represents the number of decision steps and operational checks required to reach the goal state. Urgent bins incur higher path costs due to additional reasoning steps (prediction, prioritization, routing and assignment), while safe bins terminate early with minimal cost.

12.0 Supporting the Proposed KR

The combined State Space Search operationalizes all five Knowledge Representations into a single executable reasoning framework:

1. KR #1 (Bin Fill-Level Knowledge)

Implemented through transitions from bin_fill_status to fill_level_safe or fill_level_near_full.

2. KR #2 (Waste Accumulation Prediction Knowledge)

Represented by zone_usage, high_usage_zone and bin_predicted_full, enabling early overflow prediction.

3. KR #3 (Collection Priority Knowledge)

Modeled through assess_priority_status, priority_urgent and priority_normal, integrating contextual urgency rules.

4. KR #4 (Route Optimization Knowledge)

Implemented using location states At(a) and At(c), where the system selects the shortest route.

5. KR #5 (Vehicle Assignment Knowledge)

Realized through service_request_pending, vehicle_found and vehicle_assigned, ensuring the nearest available vehicle is dispatched.

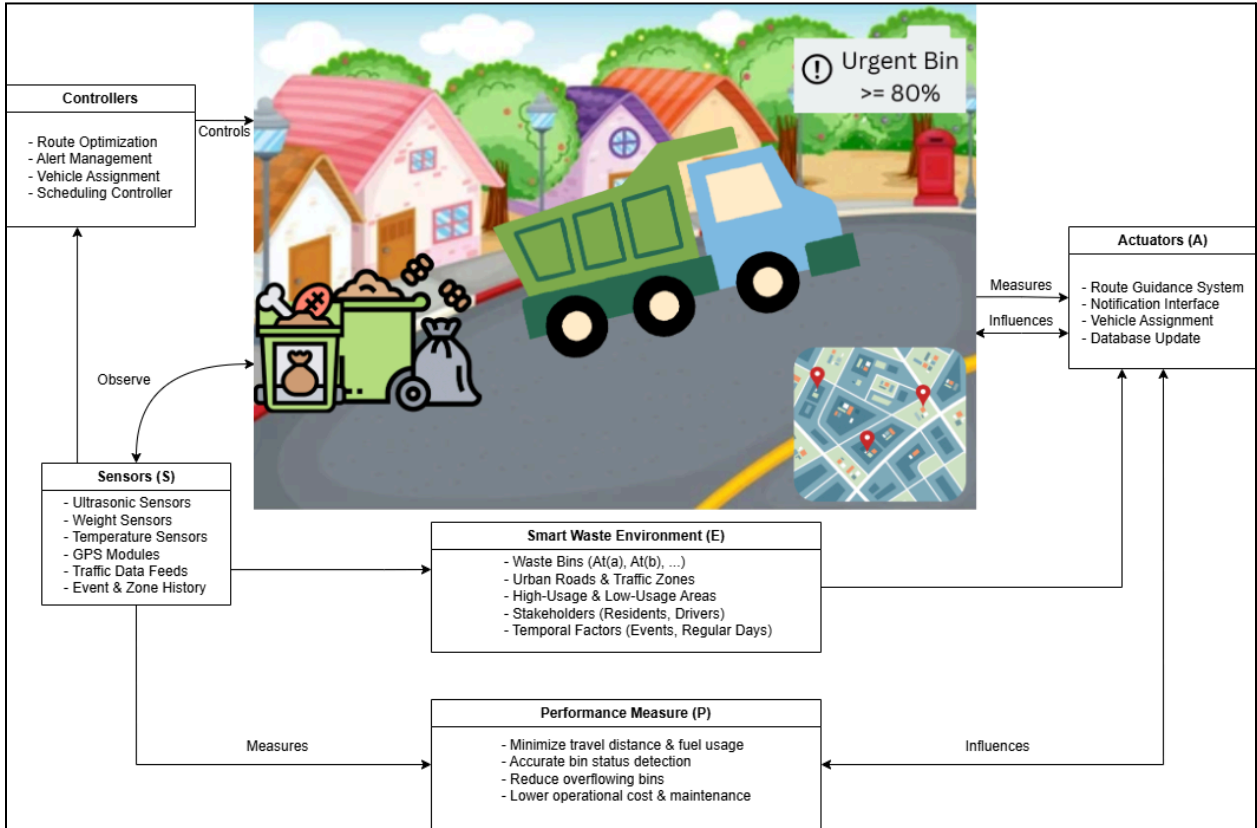
By integrating all five KRs into a unified state space, the system demonstrates how logical knowledge representations are transformed into a coherent, step-by-step AI decision process. This combined formulation ensures timely waste collection, efficient resource utilization and effective operational decision-making, fully supporting the objectives of the Smart Waste Management System.

13.0 PEAS MODEL

Agent	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Smart Waste Management Agent	Efficiency & Accuracy: Minimize total distance traveled and fuel consumption. Correct identification of Medium , Near-Full ($\geq 70\%$) , and Critical ($\geq 80\%$) bins .	Urban Layout & Zones: Roads, intersections, and traffic zones. Divided into High-Usage Zones (fast accumulation) and Low-Usage Zones .	Route Guidance System: Displaying optimized paths (shortest route) to the driver.	Hardware Sensors; Ultrasonic Sensors: To detect real-time fill height. Weight Sensors: To measure the load/mass of the waste. Temperature Sensors: To detect potential fire hazards.
	Cleanliness & Hygiene: Minimize the number of overflowing bins (collection latency), unpleasant odors, and pests.	Waste Bins: Located at various static coordinates (e.g., At(a), At(b)) with changing fill states.	Notification & Alert Interface: Issuing “Urgent Collection Requests” to the central dashboard or driver when bins reach the $\geq 70\%$ threshold.	Location & Tracking; GPS Modules: To track the real-time location of collection trucks.
	Cost & Sustainability: Reduce operational costs (labor hours, vehicle wear) and carbon emissions.	Stakeholders: Residents (waste generators) and Waste Collection Workers (drivers).	Vehicle Assignment: Electronically assigning the nearest available vehicle to an urgent bin.	Data Inputs; Traffic Data Feeds: Real-time density information. Event Calendar: To detect special event days.
	Response Time: Ensure urgent bins are detected and serviced	Temporal Factors: Event Days (holidays/festivals)	Database Updates: Logging collected bins, route completion	Zone Usage

	promptly by the nearest vehicle.	vs. Regular Days affecting waste generation rates	status, and “Safe” status after collection.	History: To predict accumulation trends.
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14.0 Relations Between Each Property



15.0 Explanation In Proof Of Concept (POC)

15.1 Performance

The agent's performance is measured by its ability to optimize waste collection logistics and maintain urban hygiene. It aims to minimize total distance travelled and fuel consumption by providing optimized route guidance to drivers. Success is also defined by the system's accuracy in identifying bin fill levels which are categorized as Medium, Near-Full ($\geq 70\%$), and Critical ($\geq 80\%$) to prevent overflows, odors, and pest issues.

15.2 Environment

The environment comprises the urban infrastructure and the stakeholders involved in waste generation and collection. This includes a network of roads, intersections, and traffic zones divided into high-usage and low-usage areas. The waste bins are located at static coordinates, such as At(a) and At(b), while temporal factors like holidays and festivals influence the rate of waste accumulation.

15.3 Actuators

Actuators allow the agent to affect the environment and manage logistics. The Route Guidance System displays optimized paths to drivers, while the Notification & Alert Interface issues urgent collection requests when bins hit the 70% threshold. Additionally, the system performs Vehicle Assignment by electronically dispatching the nearest truck and updates the Database to log completed collections and "Safe" area statuses.

15.4 Sensors

The agent perceives its surroundings through various hardware and data inputs. Ultrasonic sensors detect real-time fill height, weight sensors measure the mass of waste, and temperature sensors monitor for fire hazards. For tracking, GPS modules provide truck locations, while Traffic Data Feeds and Event Calendars help the agent anticipate road density and seasonal waste spikes.

16.0 Behaviors Of Agent To Achieve The System Goals

The Smart Waste Management Agent is a goal-based agent. Its primary goal is to maximize collection efficiency while ensuring no bins reach an overflowing state.

The agent interacts with the environment by continuously monitoring sensor data. Once active, the sensors detect the fill level of bins across various zones. The agent processes this information to compute the most efficient collection path, prioritizing bins at the critical $\geq 80\%$ threshold. For example, if a bin in a high-usage zone becomes full, the agent immediately alerts the driver and adjusts their route.

The agent also updates the central dashboard with real-time zone status and historical usage trends. When a bin reaches the 70% “urgent” threshold, the agent automatically assigns the nearest available vehicle to service that location. Throughout the process, the agent assists the driver by providing navigation that avoids heavy traffic. After collection, the agent marks the bin as “Safe”, ensuring consistent improvement in urban cleanliness and operational sustainability.

17.0 Conclusion

In conclusion, this project successfully proposed a Smart Waste Management System that uses Artificial Intelligence (AI) and Internet of Things (IoT) technologies to improve urban waste collection. By moving away from fixed schedules and introducing real-time monitoring, prediction and automated decision-making, the system is able to detect full bins early, prioritize urgent collections and assign the most suitable vehicles using optimized routes.

Through the use of Knowledge Representation, State Space Search and the PEAS model, the system demonstrates how intelligent agents can reason, plan and act effectively in a real-world environment. These components work together to reduce overflow, minimize fuel consumption, improve response time and enhance overall operational efficiency.

This project highlights the importance of IA in supporting smart city initiatives and sustainable urban management. With further development and real-world implementation, the proposed system has strong potential to improve city cleanliness, reduce costs and enhance the quality of life for residents while supporting environmental sustainability.

Link Demonstration Video -

https://drive.google.com/drive/folders/1tyqFh_1uqMfYVPPrjJa1eLHndWZ3AP56